Nicholas H. Androski

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Education

University of Michigan, Ann Arbor **August 2024-Present**
PhD in Climate and Space Physics anticipated 2029

- M.S. Climate and Space Sciences and Engineering from University of Michigan (April 2026)
- Advisor: Dr. Christiane Jablonowski

Cal Poly, San Luis Obispo **Sep. 2020 – June 2024**
Bachelor of Science in Physics with Astronomy and Mathematics Minors *GPA: 4.00*

Skills

Technical Skills

- Proficient in Python and MATLAB in data analysis and visualization including the IRAF Python framework for reducing astronomical data (Frost Summer Research; CfA Solar REU; Observational Astronomy Course)
- Unix, Fortran 90, Mathematica, C++, CERN's ROOT framework, LabView, Github and LaTeX experience
- Hardware Experience: Oscilloscopes, PMTs, circuit design, and telescopes focused on measuring the validity of physical models with proper propagation of statistical uncertainty. (Quantum Lab, Observational Astronomy, and Electronics Courses)
- Proficient in running and modifying earth system models, particularly NCAR's Community Earth System Model (CESM).

Communication

- Synthesizing and presenting technical ideas. (Production of poster and oral presentations for all of my research experiences including posters at AAS, APS Far West, AGU, and AMS meetings).
- Demystifying complex physics concepts to fellow students geared towards understanding instead of memorization. (Learning Assistant, Cal Poly [2023]; Learning Assistant Seminar Course; Undergraduate Tutoring [2022-2024]).

Research

Graduate Student Research Assistant **August 2024 – Current**
University of Michigan - Advised by Dr. Christiane Jablonowksi *Ann Arbor, MI*

- Developing idealized test cases within CESM to probe moist convective dynamics at the gray scale of convection for high resolution, km-scale, climate models.
- Working on the NSF StormSPEED project to integrate DOE's non-hydrostatic spectral element dynamical core and assess its scientific validity with a focus on idealized test cases of mesoscale convective systems.
- Using GIS tools to assess risk to environmental hazards and the inequities in who faces the worst risks within Washtenaw County as a part of the Rackham Reparative Justice Research Group.
- Contributing to the organization of non-hierarchal monthly discussions of justice topics, such as climate and energy justice, with a focus on identifying the action we can enact in the present.

Solar REU Intern **June 2023 – August 2023**
Center for Astrophysics | Harvard and Smithsonian - Dr. Kristoff Paulson *Boston, MA*

- Developed original code to extract, visualize, and analyze plasma and electromagnetic (EM) data from NASA's Parker Solar Probe (PSP) mission in order to search for EM wave activity in the Sun's corona that may contribute to the high temperatures at the crux of the coronal heating problem.
- Produced a catalog of Sub-Alfvenic intervals in PSP data which indicates when the probe is within the corona and presented the list to the PSP's SWEAP instrument science team.

Frost Research Student **June 2022 – August 2022**
Cal Poly - Dr. Elizabeth Jeffery *San Luis Obispo, CA*

- Simulated unresolved binary star systems to analyze the effectiveness of a novel technique for measuring the age of individual main sequence stars through Bayesian analysis.

- Developed a data framework in Jupyter Notebook with widgets that organized and streamlined the analysis of 100s of non-Gaussian posterior distributions.
- Presented a poster presentation at the 2023 Winter AAS meeting

Observational Astronomy Course Project — **Sept. 2022 – Dec. 2022**

- Operated the Cal Poly Observatory to take images of a pulsating variable RR Lyrae star DH Peg.
- Used IRAF software to conduct differential aperture photometery to reduce images and produce a light curve
- Analyzed the light curve with a Lomb-Scargle approach to extract a period of pulsation.
- Presented findings in a technical report and final poster presentation for the course.

Undergraduate Research Student — **March. 2023 – May 2023**
Cal Poly - Dr. Elizabeth Jeffery — *San Luis Obispo,CA*

- Standardized photometry of the metal rich open cluster NGC 6253 and used a Bayesian Analysis software (BASE-9) to constrain the cluster age by fitting the main sequence turn off age.
- Presented work as a poster at the 2024 Winter AAS meeting.

Frost Research Student — **June. 2021 – August 2021**
Cal Poly - Dr. Jennifer Klay — *San Luis Obispo,CA*

- Worked with large sample of millions of nuclear fission event data from the NIFFTE experiment with CERN's ROOT framework.
- Developed an algorithm to characterize and sort rare ternary alpha nuclear fission events.

Publications

Technical Reports

- Andorksi, N.H, Paulson, K. (2024). **Identifying Ion-Scale Waves in Parker Solar Probe Sub-Alfvénic Intervals**. *Digital Commons @ Cal Poly.* https://digitalcommons.calpoly.edu/physsp/234

Awards/Honors

- California Polytechnic State University, San Luis Obispo Philip and Christina Bailey College of Science and Math 2024 Academic Excellence Award
- Cal Poly, San Luis Obispo Department of Physics Academic Excellence Award, June 2024
- California Polytechnic State University, San Luis Obispo graduating honors *Summa Cum Laude* in June 2024
- Cal Poly, San Luis Obispo Philip and Christina Bailey College of Science and Math Dean's List: All quarters of attendance

Teaching Experiences

Learning Assistant (LA) — **March 2023 – June 2023**
General Physics II — *Cal Poly San Luis Obispo*

- Participated in a LA pedagogy seminar focused on evidence-based strategies of physics education
- Learned how to effectively guide students by avoiding direct answer-giving and instead working to unwrap their mental models with probing questions.
- Guided students in two studio (lab+lecture) style class sections of a general physics II, covering topics such as simple harmonic motion, waves, and thermodynamics.

Presentations

An Intercomparison of Convection across CESM3's Nonhydrostatic Dynamical Cores through an Idealized Model Hierarchy — **July, 15 2026**
Poster
***N. Androski**, C. Jablonowski, J. Truesdale, P. Lauritzen, J. Sun, M. Taylor*
km-scale Global Modelling Summit 2026

NSF StormSPEED: Enabling Km-scale Earth System Simulations in CESM — **April 13, 2026**
Presentation
***N. Androski**, C. Jablonowski, J. Truesdale, P. Lauritzen, J. Sun, M. Taylor*
J. Sun, M. Taylor, and the NSF StormSPEED Team
TAMU Km-Scale Workshop 2026

An Intercomparison of Idealized Squall-Lines Across CESM3's Nonhydrostatic Dynamical Cores — **July, 15 2026**
Poster
***N. Androski**, C. Jablonowski, N. Forcone, J. Truesdale, P. Lauritzen,*
J. Sun, M. Taylor, and the NSF StormSPEED Team
TAMU Km-Scale Workshop 2026

An Idealized Squall Line Test Intercomparison of Non-Hydrostatic Dynamical Cores — **Jan. 29, 2026**
Poster
***N. Androski**, C. Jablonowski*
American Metereological Society 2026

Introduction to the Hands-on component of DCMIP and the DCMIP-2025 test cases — **June, 2 2026**
Presentation
*T. Andrews, O. Hughes, **N. Androski**, C. Jablonowski*
Dynamical Core Model Intercomparison Project 2025

Workshops

TAMU High and Ultra-high Resolution Modeling of the Earth System Workshop — **April 13-15, 2026**
Participant

The World Climate Research Programme Global KM-scale Hackathon — **May 12-16, 2025**
Participant — Website

Dynamical Core Model Intercomparison Project Summer School — **June 2-6, 2025**
Organizer — Website

- Developed test case 3, the idealized squall line test, and created a pipeline for participants to run and modify the test case for themselves.
- Mentored participants within the Spectral Element (SE) group and helped to troubleshoot model difficulties and plan test case experiments.

Service

Graduate Undergraduate Student Organization (GUStO — **Fall 2025 – Current**
Student Engagement Chair — *University of Michigan*

- Organize and run events for graduate and undegraduate students within the climate and space department.
- Compiled budgets and plans for events annually.

Society of Physics Students — **Fall 2022 – Spring 2024**
President — *Cal Poly San Luis Obispo*

- Established a set of social norms for the physics community lounge in collaboration with the Women in Physics club based upon a community focused townhall in order to make the space more welcoming and safe for new students and underrepresented members of the community.
- Led weekly meetings to oversee progress and delegate tasks towards goals that promote diversity and a sense of community in the difficult journey of studying physics.
- Planned the physics department spring banquet with 100+ attendees to celebrate and honor the work of faculty and students